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Course 608

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From Root Architecture to Network Resilience: A Quantitative Framework For Root-Inspired Topology and Decentralized Failure Recovery

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Abstract

Modern computer networks must continue operating despite component failures, link disruption, congestion, and targeted attacks. The topology of the network plays a major role in failure propagation, and resilience is also dependent on the ability of a system to recover from the failures [1], [2]. Plant root systems provide a potentially useful biological model because they form distributed branching architectures and adjust their development depending on the local environmental conditions [3], [4]. Particularly, root architecture can be quantified [5]. Sultan et al. studied 1,645 3D root structures across four different species and found consistent spatial-density patterns within populations [5].

The research proposes a new computational paradigm that uses measurable features of roots (branching structures, spatial density, and adaptation) for generation and recovery of networks. Root-inspired networks will be compared with small-world, scale-free, and hierarchical topologies under random node, targeted node, edge, and cluster failures. An adaptive version will use local network information to reroute traffic and introduce selective alternative connections in case of degradation. Structural resilience will be measured using connectivity, efficiency, fragmentation, and path-based metrics, while packet-level experiments will evaluate delivery ratio, latency, throughput, recovery time, and control overhead. Finally, ablation tests will identify which root-derived mechanisms contribute to resilience. The study will therefore test which biological principles can be transferred to network engineering, under which failure conditions, and at what cost, rather than assuming that biological inspiration is inherently superior.

Introduction

Large distributed networks are subject to hardware failures, communication link disruption, congestion, physical damage, and malicious attacks. Previous researches has shown that network architecture can determine whether a system remains connected after random failures or becomes highly vulnerable when structurally important nodes are removed [1]. Recent studies have suggested that resilience is a multi-dimensional attribute that involves resistance to disruption as well as the ability to recover from it[2].

Plant roots provide an interesting model of a distributed architecture. Root systems consist of branching structures whose architecture is influenced by developmental and environmental signals [3], [4]. Fitter showed that root systems can be described through quantitative architectural and topological properties, such as branching organization, link length, and spatial exploration [6]. Some researchers have applied mathematical topology to model plant branching [7], while Sultan et al. demonstrated that the spatial density of hundreds of thousands of root architectures can be explained using a common statistical framework [5]. These findings suggest that network designs that are inspired by roots can be based on quantitative biological traits and not just on similarities alone.

Bio-inspiration in networking has shown that there are biological mechanisms that have been able to inspire decentralized computational mechanisms [8]. Tero et al., for example, famously used the growth patterns of Physarum (slime mold) to build network design rules that neatly balance transport efficiency, infrastructure costs, and fault tolerance [9]. The root-inspired network has extended this bio-inspired approach to plant root architecture.

Research Aim, Question and Hypotheses

Aim: To determine whether quantifiable properties of plant-root architecture and adaptive branching can be incorporated into graph-generation methods and decentralized self-recovery techniques to increase the robustness of distributed computing networks while maintaining efficient communication and reasonable costs.

Research Question:

Central Question: What specific quantifiable properties of plant root system, when translated into graph generation algorithms and local repair rules, improve the robustness of distributed networks in case of random, targeted, edge, and clustered attacks compared to small-world, scale-free, and hierarchical networks, and what are the related structural and communicational overheads?

Sub-Questions:
1. What measurable traits of plant-root architecture can be translated into computational algorithms for the generation of network topologies?
2. How will root-inspired topologies fare in dealing with random, targeted, edge, and clustered failures compared to other well-known network models?
3. Is it possible for decentralized adaptation using local network information to improve recovery from failure and/or congestion?
4. What is the relationship between resilience, redundancy and cost?
5. What are the specific root-based features which improve recovery?

Hypotheses:

1. Root-inspired networks will possess structural differences compared to traditional network models.
2. Under certain types of failure and/or equal node and edge budget, root-inspired topology will maintain connectivity and/or efficiency.
3. Adaptive root inspired network will recover from failure better than Static network.
4. Local adaptation will be able to recover from failure without needing global knowledge of the network.
5. Redundancy will increase resilience at the cost of increased cost.

Literature Review

The research on complex networks shows that network topology affects the tolerance to failures. Albert, Jeong, and Barabási have shown that the results of random perturbations and targeted attacks differ depending on the network topology [1]. Artime et al. have stressed that robustness and resilience should be measured by several criteria [2].

The root research forms the biological background for the proposed study. The branching pattern of roots is known to have plasticity in response to environmental and developmental conditions [3]. Fitter has developed a topology-based model to compare root architecture and to define the trade-off between exploration, transport efficiency, and structural costs [6]. Magalhães and

Seifert have then described the branching topology and self-similarity of plants' root systems [10]. Li et al. also showed that persistent homology can be applied to plant branching [7]. More specifically, Sultan et al. have analyzed 1,645 root architectures four different species and have revealed the population-similar spatial density function in 3D space [5].

Literature Gap: Even with the development seen thus far, a disconnection remains between root structure quantification and network resilience. There have been efforts to explore the applicability of biological concepts in network design before; [8], [9] show how effective the biological approach is in designing distributed networks. However, root inspired network explores the possibility that measurable root structures and selective redundancy can be mapped to network structure and distributed network recovery.

Methodology

The proposed framework relies on two core building blocks:

Root-Inspired Topology Generator: Graph generation rules will be created based on architectural parameters found in scientific papers concerning the roots, which are, branching probability, branching order, spatial density, branch length, and local redundancy. The generated network will take the form of a graph G=(V, E), where nodes will represent computational/network endpoints and edges will represent communication links.

Decentralized Failure Detection and Repair: Nodes will use limited local knowledge such as the availability of neighboring links, queue state, and local connectivity. Once the network degrades past certain thresholds, traffic will be re-routed and possibly some alternative links will be formed. This adaptability algorithm will be based on the concept of local biological reactions, and not on any sort of plant hormone simulations.

Key variants that will be compared are:

1. Static Root inspired Network: a root-based network topology without adaptive capabilities.
2. Adaptive Root inspired Network: a root-based network topology with local detection and selective redundancy.

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Other comparison network topologies will be hierarchical/tree, Erdős-Rényi random, Watts-Strogatz small world, and Barabási-Albert scale free. Network topologies will be chosen with similar numbers of nodes, edges, and average degrees to prevent an artificial advantage of resiliency based on resource imbalance.

Experimental Setup

All networks will then be exposed to increasing levels of random node failure, targeted node failure, random edge failure, and cluster failure. Where possible, cascading overload tests will also be carried out. The generation and analysis of the graphs will be done using Python/NetworkX, while packet-level tests will be done using ns-3. Ablation studies will involve

removing one characteristic at a time, viz., branching, clustering, selectivity, adaptation, and recovery.

Data Collection and Data Analysis

For this research, two types of data will be gathered:

1. Structure: largest component, global efficiency, average path length, fragmentation, algebraic connectivity, and extra links.
2. Communications: delivery ratio, latency, throughput, packet loss, queue congestion, recovery time, and routing overhead.

Many different networks and random seeds will be tested in each situation. Results will include failure-performance curves, variation, confidence intervals, and effect sizes, determined through suitable statistical analyses and multiple comparison corrections. In addition, a cost-resilience study will analyze how resilience relates to edge number, edge cost, cost of recovery calculations, and communication overhead.

Expected Outcome
This research does not necessarily assume that a root-inspired network will outperform all existing architectures. The topology that draws inspiration from root systems might have some benefits during localized failures but may not perform well in the case of attacks on the system. An adaptive strategy for recovery would improve the performance but increase the control overhead. Certain biological traits may not bring any noticeable gain.

Hence, the major expected contribution of this study will be the analysis of mechanisms used in biological systems. Instead of assuming that the root system is a better model than existing ones, the goal will be to discover which properties can be successfully translated into network systems.

Potential Limitations and Roadblocks
The biological analogy is incomplete. Roots move water and nutrients, but computer networks move information. Graph modeling cannot fully capture the complex biochemical processes at play in plants. Thus, biological characteristics are modeled as abstractions in engineering terms rather than biological simulations. There are further issues related to the size and traffic, failure modes, and parameters used in the simulation. Such issues will be addressed using sensitivity analyses, matching comparison networks, different random seeds, and ablation tests. The performance of the simulation does not imply anything about the performance of the physical network.

Timeline
The research will be conducted over 18 weeks. It can be completed primarily with open-source software and publicly available biological data.

Week 1-2: Literature review and root-parameter extraction
Week 3-5: Root-inspired topology generator development and validation

Week 6-7: Baseline Network Generation
Week 8-9: Failure Experiments
Week 10-12: Adaptive root-inspired network development Week 13-14: ns-3 packet-level experiments
Week 15-16: Statistical Analysis and ablation
Week 17-18: Final analysis, writing and reproducibility package

Conclusion

Root-inspired network provides the quantitative connection between the architecture of plant roots and resilient computer networks. Past research shows that the roots have measurable properties of branching and space [3]-[8], and network theory proves that topology and adaptivity greatly influence resilience [1], [2], [8], [9]. This paper does not assume that the roots represent an ideal system but instead attempts to find out experimentally whether there is any principle of root architecture that would improve network resilience in particular conditions of failures and at particular costs of structure and communication.

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